

Ashkan Pirmani

Applied machine learning · Real world evidence · Health data held in many places

NOW Postdoctoral researcher, KU Leuven and Hasselt University. Innovation Mandate, a Flemish government grant, with Johnson & Johnson Innovative Medicine as industry partner.

PHD Dual PhD, 2024. Electrical engineering at KU Leuven, biomedical sciences at Hasselt University.

EVIDENCE	45.5% more patient records in a worldwide COVID-19 study, through the route I built	26,246 patients in my study of models that adapt to each country	30+ countries whose registries and consortia I have worked with
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Experience

2024 to now

Postdoctoral researcher

KU Leuven, Department of Electrical Engineering, and Hasselt University, Data Science Institute. Belgium.

- Lead FLkit, an open guide for starting studies across centers that keep their own data. It went from an internal need to a public resource, designed to help teams get started on their own.
- Wrote the Innovation Mandate proposal and defended it. Awarded in 2026, with Johnson & Johnson Innovative Medicine as industry partner.
- Took the disability progression study to publication as first author, in npj Digital Medicine (2025).
- Designed FLoRank, a way for each site to keep a small private part of a shared model. Prototype. Awarded an ECTRIMS personal grant in 2025.
- Retired FL4E and wrote up why it was not reused, for Medical Informatics Europe 2026.
- Organizing team of OHDSI Europe 2025. Invited talks at Roche, the Flower community and the North American Society of Artificial Intelligence in Multiple Sclerosis.
- Coordinate with data holders, clinicians and legal teams in more than thirty countries. Supervise master's students.

2020 to 2024

Doctoral researcher

KU Leuven and Hasselt University.

- Built the federated route into the worldwide multiple sclerosis and COVID-19 data collection, so registries that could not send data could still take part. It added 3,527 records. First author, JMIR Medical Informatics (2023).
- Built FL4E, a framework for federated studies, and the degree of federation. First author, JMIR Formative Research (2024).
- Ran the first systematic comparison of personalized federated learning for two year disability progression in multiple sclerosis, on 26,246 patients.
- Taught in the Data Science in Healthcare course at Hasselt University, and began supervising master's theses in 2021.
- Presented at ECTRIMS and Medical Informatics Europe in 2024.

2017 to 2019

Research assistant, Decision Support Systems Lab

IranDoc, Iranian Research Institute for Information Science and Technology, Tehran.

- Agent based and system dynamics simulation of the national parcel network, searched with a genetic algorithm for a better set of hubs. Best scenario: 15% lower projected transport cost, in simulation.
- Delivered as a decision support dashboard and a policy analysis for senior executives.

Education

2020 to 2024

Dual PhD

Electrical engineering, KU Leuven. Biomedical sciences, Hasselt University. Defended November 2024.

- Thesis: From Centralized to Federated. The Journey of Data in Healthcare.
- Supervisors: Yves Moreau (KU Leuven), Liesbet M. Peeters (Hasselt University). Co-supervisor: Niels Hellings.

2016 to 2019

MSc, Industrial Engineering

Socio-economic systems. Kharazmi University, Tehran.

2015 to 2017

MBA, Quality Engineering

Tose'e Institute, Tehran.

Recognition

2026

Innovation Mandate

Flanders Innovation and Entrepreneurship, with Johnson & Johnson Innovative Medicine as the industry partner. Drug safety research on health records and insurance claims.

2025	ECTRIMS personal grant for scientific quality For FLoRank, at ECTRIMS 2025, Barcelona.
2019	Fully funded PhD scholarship KU Leuven and Hasselt University, through to the defense in 2024.

Selected contributions

45.5% more records

Designed and built the federated route into a worldwide COVID-19 and multiple sclerosis study. The four registries counted in the paper added 3,527 records to the 7,757 already collected, and a fifth joined later. Analyses of the combined data set informed worldwide COVID-19 advice for people with multiple sclerosis.

JMIR Medical Informatics 11(1):e48030, 2023

Personalized to each country

Personalizing a federated model to each country improved AUC-PR by 31% over the same method without personalization, and ROC-AUC by 3.8% over one model trained on all records pooled together. The best model uses AdaptiveDualBranchNet, an architecture I designed. Simulated federation on records from 26,246 patients and 146 centers. First author, in a collaboration of 73.

npj Digital Medicine 8(478), 2025

The degree of federation

Each center chooses whether its records leave. On a public benchmark, letting half the centers send their records kept 97.5% of the fully federated score.

JMIR Formative Research 8:e55496, 2024

A public resource, maintained

FLkit: 39 pages, eleven ways in by role, seven worked project stories, 29 contributors. Supported by ELIXIR Belgium. Its use has not been evaluated yet.

arXiv 2606.23500, 2026

Teaching and supervision

2022 to 2024	Data Science in Healthcare Teaching member, Healthcare Engineering program, Hasselt University. Two academic years.
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2021 to now

Master's thesis supervision

Seven master's theses in artificial intelligence, KU Leuven.

- Predicting admission to care. Federated, pooled and local data compared.
- How splitting the data changes what the model learns.
- Federated boosting in semi-supervised learning.
- Contrastive learning for federated models.
- Personalized models. Architectures and parameter exchange.
- Fine-tuning a global federated model for local performance.
- Early disease detection from shallow whole genome sequencing of cell-free DNA.

Methods and systems**Methods**

- Federated learning and federated analysis
- Personalization when sites hold different patients
- Privacy-preserving machine learning
- Survival analysis and classification on clinical data

Data

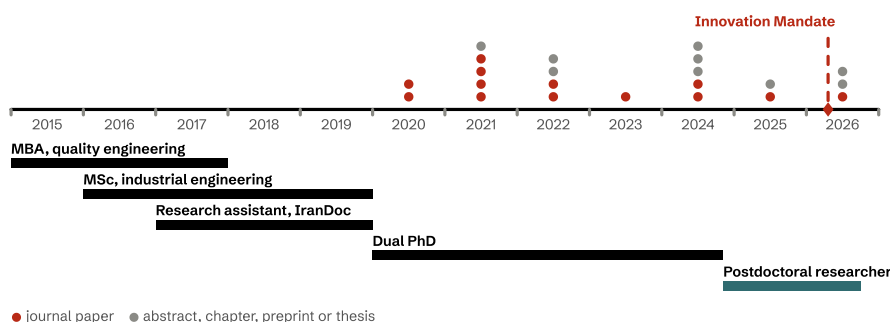
- Real world and observational data
- Cohort building and phenotype definition across registries
- Registry harmonization and data quality
- Data feasibility: whether a question can be answered at all
- OMOP common data model, through OHDSI
- FAIR data practice

Tools

- Python, PyTorch, Flower, scikit-learn, pandas
- R and SQL
- Docker, Kubernetes, continuous integration

Networks

- OHDSI Belgium, and the organizing team of OHDSI Europe 2025
- ELIXIR Belgium
- The MSBase registry and the MS Data Alliance

Selected publications

● journal paper ● abstract, chapter, preprint or thesis

Roles as bars, and every entry in the publications list as a dot in its year.

2026**Development and Design of FLKit: A Structured Onboarding Toolkit for Federated Learning in Health and Life Sciences**

arXiv preprint arXiv:2606.23500. Pirmani, Vermeulen, Vinterhalter, Geys, Faes, Ali, et al.

2026**Good for All, Not Good Enough for One: Reuse Dilemma in Federated Learning***Studies in Health Technology and Informatics.* Pirmani, Moreau, Peeters**2025****FLoRank: adaptive structured personalisation for federated prediction of disability progression in multiple sclerosis***ECTRIMS 2025, Multiple Sclerosis Journal 31(3 suppl).* Pirmani, Moreau, Peeters**2025****Personalized federated learning for predicting disability progression in multiple sclerosis using real-world routine clinical data***npj Digital Medicine.* Pirmani, De Brouwer, Arany, Oldenhof, Passemiers, Faes, et al.**2024****Accessible ecosystem for clinical research (federated learning for everyone): development and usability study***JMIR Formative Research.* Pirmani, Oldenhof, Peeters, De Brouwer, Moreau**2023****The Journey of Data Within a Global Data Sharing Initiative: A Federated 3-Layer Data Analysis Pipeline to Scale Up Multiple Sclerosis Research***JMIR Medical Informatics.* Pirmani, De Brouwer, Geys, Parciak, Moreau, Peeters**2022****Updated results of the COVID-19 in MS global data sharing initiative: anti-CD20 and other risk factors associated with COVID-19 severity***Neurology: Neuroimmunology & Neuroinflammation.* Simpson-Yap, Pirmani, Kalincik, De Brouwer, Geys, Parciak, et al.**2021****Associations of disease-modifying therapies with COVID-19 severity in multiple sclerosis***Neurology.* Simpson-Yap, De Brouwer, Kalincik, Rijke, Hillert, Walton, et al.